MARINE INTERNAL COMBUSTION ENGINES (MME 467)

EXHAUST EMISSIONS AND CONTROL

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IMO Marine Engine Regulations

- **International Maritime Organization (IMO)** is an agency of the United Nations that has been formed to promote maritime safety.
- There are two conventions involved: The International Convention for the Safety of Life at Sea (SOLAS) and The International Convention for the Prevention of Pollution from Ships (MARPOL).
- The structure of the IMO includes two main committees: the **Maritime Safety Committee (MSC),** which covers safety and security-related matters; and the **Marine Environment Protection Committee (MEPC)**, which addresses environmental issues under IMO's remit.
- SOLAS requires ships to undergo a regular cycle of special surveys at five-yearly intervals with an interim survey taking place between two and 3 years after each special survey.
  - ✓ It is usual at these surveys for major repairs and overhauls to take place, but the SOLAS rules only require that the main machinery should be in a satisfactory condition and fit for purpose.
- It is the MARPOL convention that impacts most engine development and operation. The convention has six annexes distinguished by Roman numerals. Annex VI is devoted to emissions of air from engines and boilers, and this is where most regulation of engines emanates from.

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Exhaust Gas Emissions

- Exhaust gas emissions from marine diesel engines largely comprise nitrogen, oxygen, carbon dioxide, and water vapour, with smaller quantities of carbon monoxide, oxides of sulphur (SOx), and nitrogen (NOx), partially reacted and non-combusted hydrocarbons (HC), and particulate material.
- For a vessel with a typical two-stroke engine using fuel with a sulphur content of 3%, the results of the combustion process contain about 600 ppm of SOx and 1500 ppm of NOx.
- In addition, for every tonne of fuel consumed more than 3 tonnes of CO_2 is released. There are also unburnt hydrocarbons and small amounts of particulates which are considered harmful to health.
- Currently, fuel with a sulphur content above 0.5% is prohibited unless the vessel is fitted with an exhaust gas cleaning system. In such a vessel, the SOx is converted to sulphates which are a normal component of seawater.
- **Nitrogen oxides (NOx)** generated thermally from nitrogen and oxygen at high combustion temperatures in the cylinder are of special concern since they are believed to be carcinogenic and contribute to photochemical smog formation over cities and acid rain (and hence excess acidification of the soil).

✓ Internal combustion engines primarily generate nitrogen oxide but <10% of that oxidizes to nitrogen dioxide the moment it escapes as exhaust gas.

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Exhaust Gas Emissions

- **Sulphur oxides (SOx)** produced by the oxidation of sulphur in the fuel have an unpleasant odour, irritate the mucous membrane, and are a major source of acid rain (reacting with water to form sulphurous acid).
- **Hydrocarbons (HC)** created by the incomplete combustion of fuel and lube oil, and the evaporation of fuel have an unpleasant odour, are partially carcinogenic and smog-forming, and irritate the mucous membrane (emissions, however, are typically low for modern diesel engines).
- **Carbon monoxide (CO)** resulting from incomplete combustion due to a local shortage of air and the dissociation of carbon dioxide is highly toxic but only in high concentrations.
- **Particulate matter (PM)** is a complex mixture of inorganic and organic compounds resulting from incomplete combustion, partly unburned lube oil, thermal splitting of HC from the fuel and lube oil, ash in the fuel and lube oil, sulphates, and water.

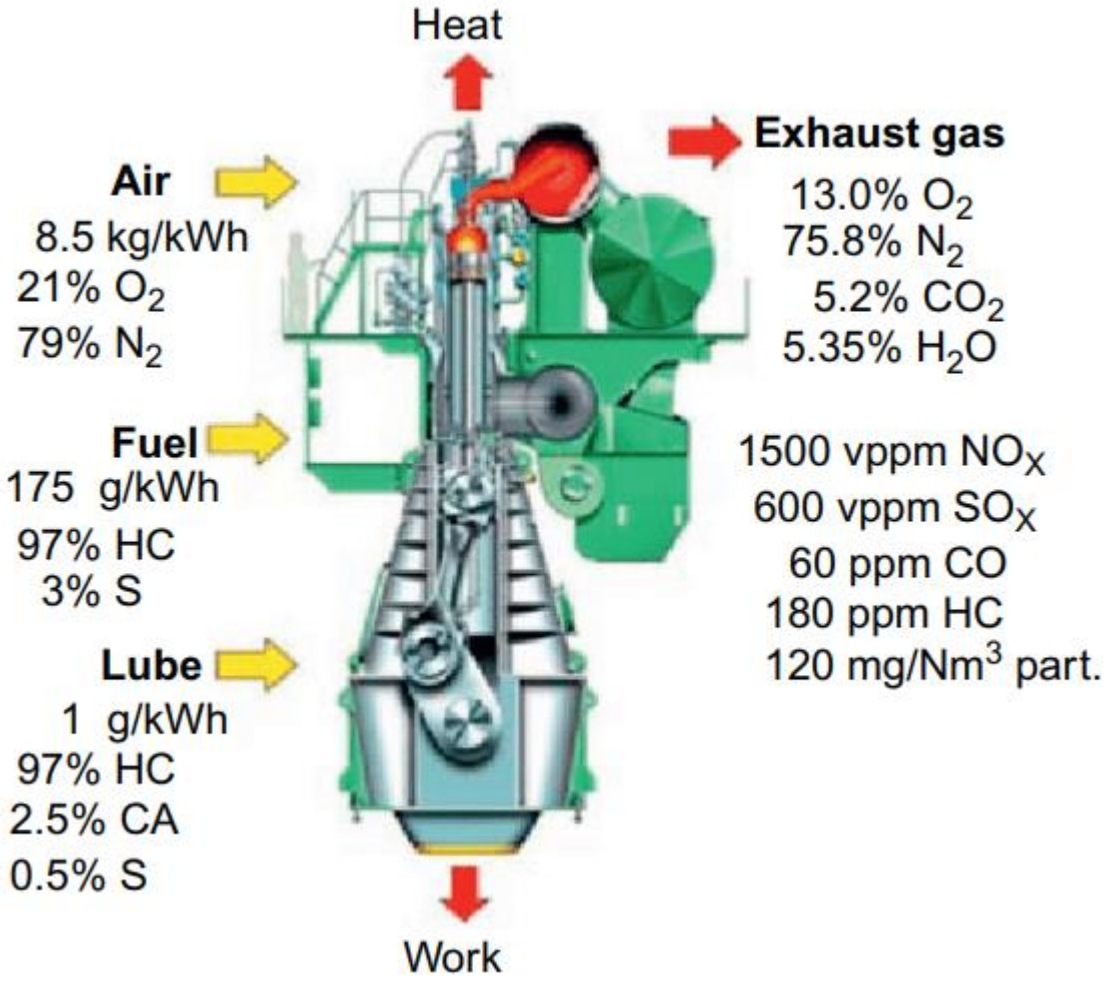
✓ More than half of the total particulate mass is soot (inorganic carbonaceous particles), whose visible evidence is smoke.

✓ Soot particles (unburned elemental carbon) are not themselves toxic, but they can cause the build-up of aqueous HC, and some of them are believed to be carcinogens. Particulates constitute no more than around 0.003% of engine exhaust gases.

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Regulating NOx Emissions

- The global approach to NOx emission controls was taken by the IMO, whose Annex VI (Regulations for the Prevention of Air Pollution from Ships) to MARPOL 73/78 was adopted at a diplomatic conference in 1997.
- IMO's current maximum allowable NOx emission levels depend on the speed category of the engine and range from 17 g/kWh for engines of speed <130 rev/min to 9.84 g/kWh for engines of speed >2000rev/min.
- Much tougher curbs on NOx and other emissions are set by regional authorities such as California's Air Resources Board, and Sweden has introduced a system of differentiated port and fairway dues, making ships with higher Nox emissions pay higher fees than the more environment-friendly tonnage of a similar size.



Typical two-stroke engine combustion products

Regulating NOx Emissions

- Amendments agreed by the IMO in 2008 set progressively tighter NOx emission standards for new engines, depending on the **date of their installation and engine speed**.
- Tier I and Tier II limits are global, while the Tier III standards apply only in NOx Emission Control Areas (ECA).

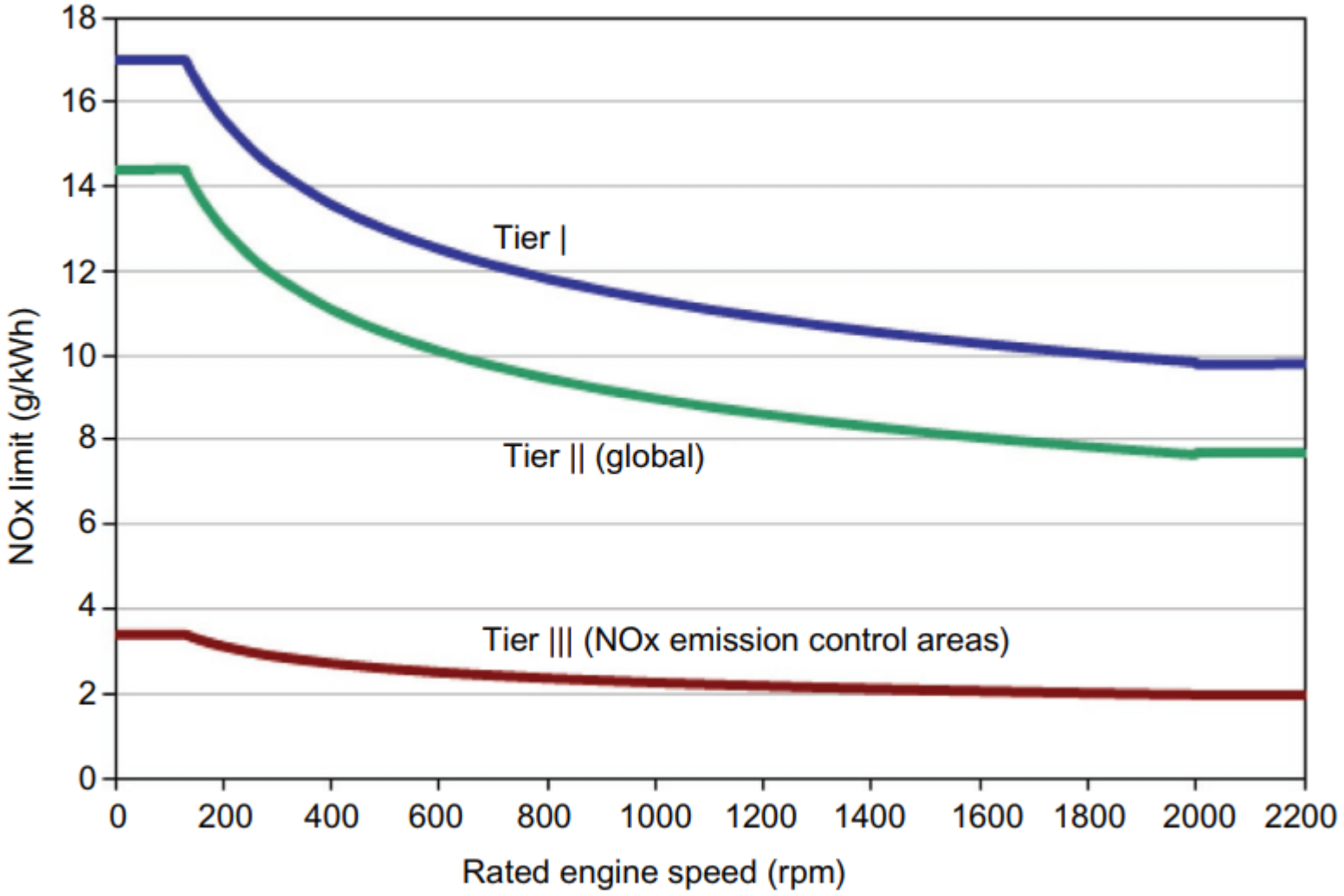
MARPOL Annex VI NOx emission limits

		NOx limit (g/kWh)		
Tier	**Date**	$n < 130$	$130 \leq n < 2000$	$n \geq 2000$
Tier I	2000	17.0	$45 \times n^{-0.2}$	9.8
Tier II	2011	14.4	$44 \times n^{-0.23}$	7.7
Tier III	2016[a]	3.4	$9 \times n^{-0.2}$	1.96

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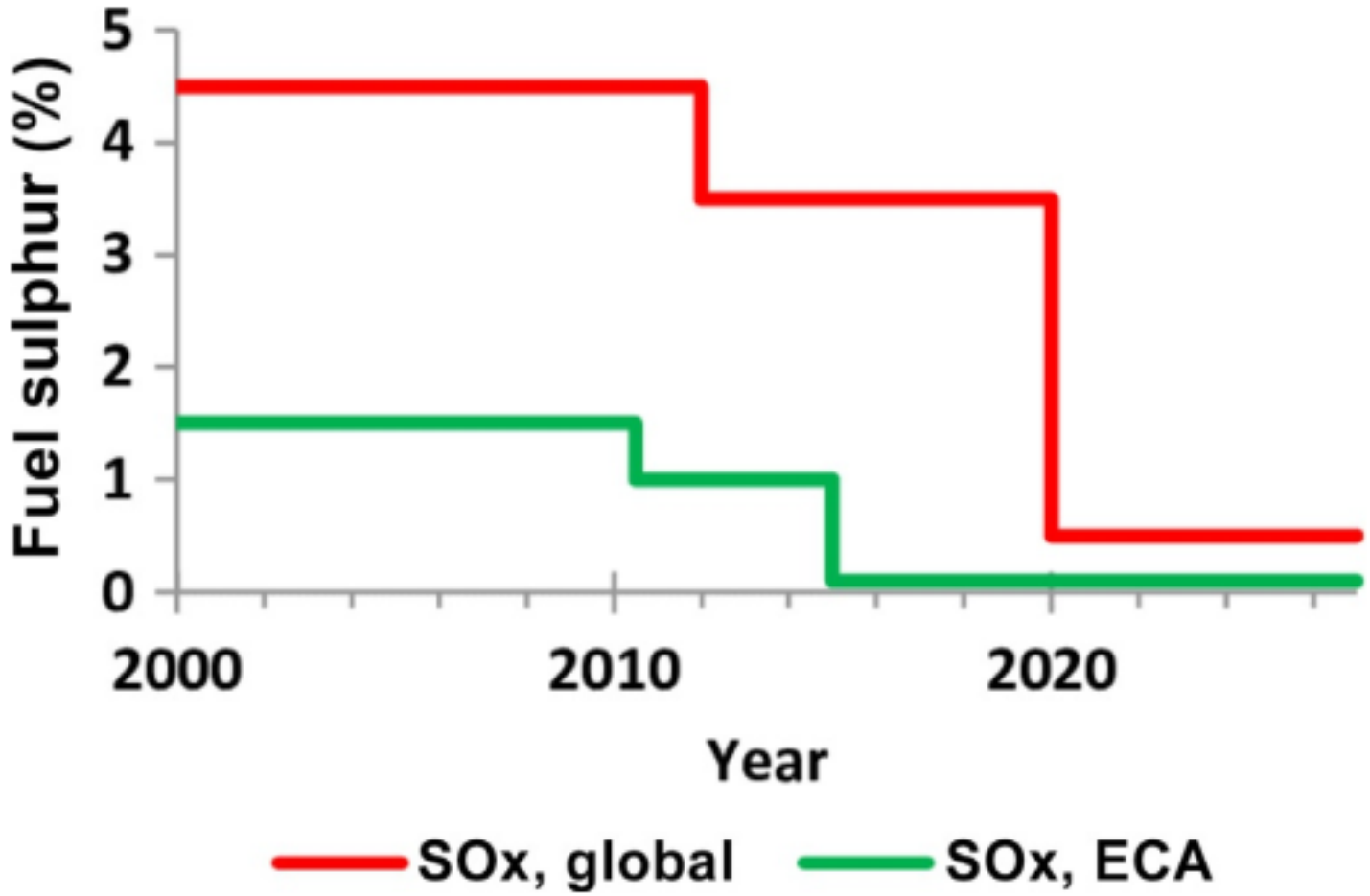
Regulating NOx Emissions

- With stricter controls planned by the IMO, the reduction of NOx emissions remains a priority for engine designers whose concern is to secure environmental acceptability without compromising the impressive gains in engine fuel economy and reliability achieved in recent years.



Regulating SOx Emissions

- After setting NOx reduction targets, the IMO next set about tackling pollution by SOx.
- Since the formation of SOx depends on the fuel rather than the engine, the IMO regulations place limits on the sulphur content of the fuel.
- First introduced in 2005, the limits were subject to a gradual tightening in the same way as the NOx regulations.
- Two limits on permissible sulphur levels were set, one for ECAs and another for areas outside of any ECA.



Regulating Other Emissions

- Because much of the PM in exhausts is related to the sulphur content, the lower permitted level of sulphur in fuels goes someway to reducing it.
- Ships equipped with wet scrubber systems will release far less PM into the atmosphere because much of it will be trapped in the wash water and form the sludge that is disposed of later.
- Some 6% of the exhaust gas emissions from marine engines is carbon dioxide. Although not itself toxic, carbon dioxide contributes to the greenhouse effect (global warming and climate change). This gas is an inevitable product of the combustion of all fossil fuels, but emissions from diesel engines are the lowest of all heat engines.
- A lower fuel consumption translates to reduced carbon dioxide emissions since the amount produced is directly proportional to the volume of fuel used and, therefore, to the engine or plant efficiency.
  - ✓ As a rough guide, burning 1t of diesel fuel produces approximately 3.25t of carbon dioxide.
- Under international agreements, such as the Kyoto Protocol and the European Union's accord on greenhouse gases, many governments are committed to substantial reductions in total emissions of carbon dioxide.

Regulating Other Emissions

- There are currently no mandatory limits on carbon dioxide emissions from engines, although the IMO does have a mandatory system in place for the measurement of the amount produced by ships.
- There is also a mandatory system in place whereby newbuild ships are obliged to improve their efficiency relative to a determined baseline. **This is known as the Energy Efficiency Design Index (EEDI)**.
- Under EEDI rules, ships are measured to a complex formula that attempts to calculate the amount of carbon dioxide produced per tonne/mile of work done using the carrying capacity of the ship.
- In 2018, The IMO adopted an ambitious target of decarbonization of the shipping industry. The aim is to reduce total emissions from shipping by 50% in 2050 and to reduce the average carbon intensity by 40% in 2030 and 70% in 2050, compared to 2008.

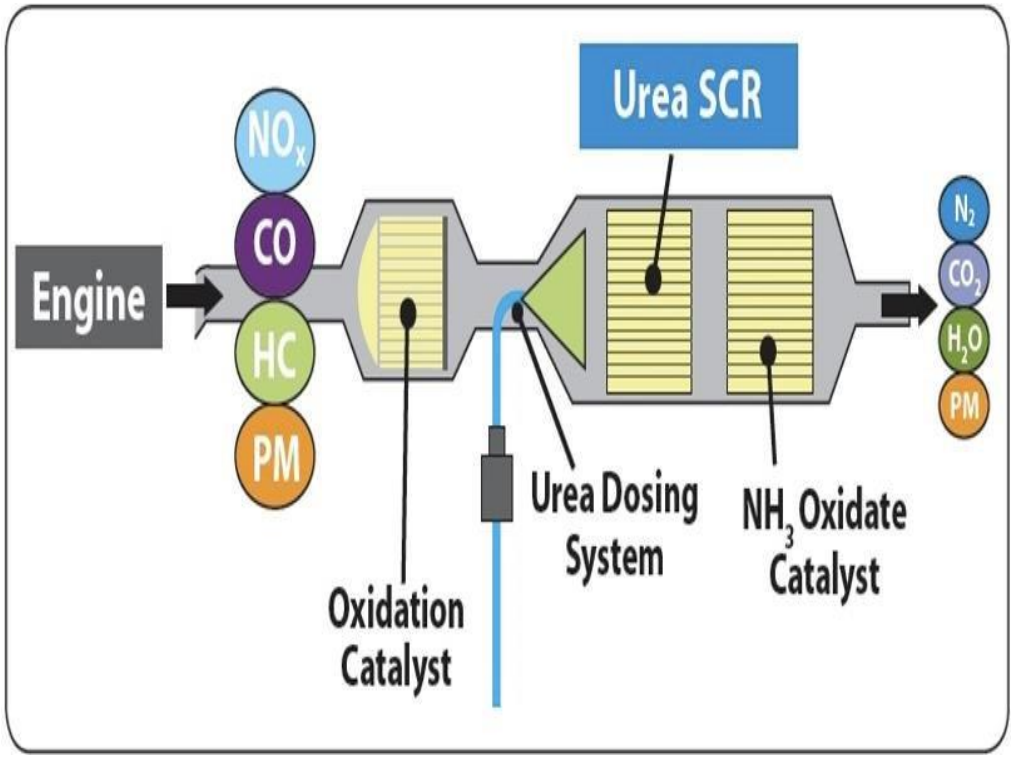
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Controlling NOx Emissions

- Two main approaches can be pursued in reducing NOx emission levels.
- **Primary (in-engine)** measures aimed at reducing the amount of NOx formed during combustion by optimizing engine parameters with respect to emissions (valve timing, fuel injection, and turbocharging). Emission levels can be reduced by 30%–60%.
- Primary NOx reduction measures can be categorized as follows:
  - ✓ Water addition: either by direct injection into the cylinder or by emulsified fuel
  - ✓ Optimized combustion-chamber geometry: for superior combustion and avoidance of peak temperatures that produce NOx formation.
  - ✓ Altered fuel injection: retarded injection, rate-modulated injection, and a NOx-optimized fuel spray pattern, Miller cycle.
  - ✓ Combustion air treatment: supercharging, turbo-cooling, intake air humidification, exhaust gas recirculation (EGR), and selective noncatalytic reduction.
- **Secondary** measures designed to remove NOx from the exhaust gas by downstream cleaning techniques such as selective catalytic reduction (SCR). Emission reductions of over 95% can be achieved.

Selective Catalytic Reduction (SCR)

- In an SCR system, the exhaust gas is mixed with ammonia (preferably in the form of a 40% solution of urea in water) before passing through a layer of special catalyst at a temperature between 290°C and 450°C.
- The lower limit is mainly determined by the sulphur content of the fuel: at temperatures below 270°C ammonia and SOx will react and deposit as ammonium sulphate.
- At excessive temperatures, the catalyst will be degraded depending on the exhaust gas temperature; therefore, reliable SCR operation calls for fuels with low sulphur contents.
- Urea decomposes into ammonia and carbon dioxide on injection into the hot exhaust gas stream.
- The ammonia reduces NOx to harmless gaseous waste products, water and nitrogen; and parts of the soot and HC in the exhaust are also removed by oxidation in the SCR process reactor.



Controlling SOx Emissions

- The principal approaches to controlling SOx emissions include the use of low-sulphur fuel; reduction or removal of sulphur in the feed; use of appropriate combustion technologies; and emissions control technologies such as sorbent injection and flue gas desulfurization (FGD).
- **Sorbent injection** involves adding an alkali compound to the coal combustion gases for reaction with the sulphur dioxide. Typical calcium sorbents include lime and variants of lime. Sodium-based compounds are also used. Sorbent injection processes remove 30–70% of sulphur oxide emissions.
- **Flue gas desulfurization** may be carried out using either of two basic FGD systems: regenerable and throwaway. Both methods may include wet or dry processes. Currently, more than 90% of utility FGD systems use a wet throwaway system process.
- Wet processes are highly efficient and can achieve SOx removal of 90% or more.
- With dry processes, solid sorbents capture the sulphur oxides. Dry systems have 70–90% sulphur oxide removal efficiencies and often have lower capital and operating costs, and lower energy and water requirements.

Table 1. Comparison of SO_x Emissions Control Systems

System	*Percent SO_x reduction*	*Capital cost ($/kilowatt)*
Sorbent injection	30–70	50–100
Dry flue gas desulfurization	70–90	80–170
Wet flue gas sulfurization	>90	80–150

Source: Kataoka 1992.


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